

1 Throughput Batch Processing

Overview

Throughput batch processing is a special application case in material and production logistics. In this case, an output batch is to adopt the visible batch number of the input batch used.

Usage/Procedure

In contrast to normal input and output batch processing in an operation requiring batch management, in throughput batch processing an input batch with a batch number is used and this batch number is handed down to the output batch.

Throughput batch processing is used for instance if the condition of a batch/material might change after a work stage but the external batch number (e.g. on a label) is to be retained.

As a consequence, no new material will occur when the output batch is changed and the batch number can even be handed down through several process steps.



The operator uses an input batch on a machine/operation. The output batch number is not changed and hence remains identical to the input batch number.

The so-called throughput batch number (throughput batch number/external batch number) therefore remains identical. Within the system, however, a unique batch number (HYDRA batch number/internal batch number) is still used and/or generated for each production level/after each output batch change, since every object within the system is unique.

If, for instance, a selection according to throughput batch numbers is made in the batch data overview, several entries with different internal batch numbers will be obtained for each throughput batch number; these internal batch numbers ensure unambiguity within the system and consequently allow for a historic observation of the "throughput batch".

This means that a total of three different statuses are considered for the process description:

- Status 1: The batch as an input batch (prior to logon)
- Status 2: The batch status on the machine (running on OP)
- Status 3: The batch as an output batch (after logoff)

The batch as an input batch (prior to logon):

Prior to logging on the batch, the information on the batch (e.g. in the batch data overview) reads as follows:

- the throughput batch number *dllosnr* "**DLLOS01**" is identical to the batch number *losnr* "**DLLOS01**"
- the batch status is "**F**" (free)
- the throughput batch flag *dll_kennz* is "**N**"

The batch status on the machine (running on OP)

When an operation is logged on to the terminal, the material (the material number) is used according to the component list from the OP in order to determine whether it is treated as a throughput batch on the basis of the material type entered for this material. This procedure also enables logging on the throughput batch number on this operation. It is not possible to log on more than one input batch/material as "throughput batches".

After logging on the batch (with throughput batch number/external batch number) on the operation/machine, the information on the batch reads as follows:

- the batch status is changed to "**L**" (running)
- the batch receives the throughput batch flag *dll_kennz* "**E**" (throughput input batch running)

After logging off the output batch on the OP/machine, a new internal object is created in the system to take over the throughput batch number of the input batch.

The information on the new output batch then reads as follows:

- a new batch with *losnr* **PR41E9C114** and *dllosnr* **DLLOS01** is created in the output buffer
- the status of the new batch is changed to "L" and
- the batch receives the throughput batch flag "G".
- The quantity on the new batch is identical to the quantity of the original input batch.

Functionality: Console Evaluation of Throughput Batches

An operator can identify and trace a created batch via the external batch number (e.g. in order to be able to forecast when a specific batch will leave production).

Material Movement Functionality

An operator can call up the material movements (goods issue/goods receipt) for an input batch/output batch.

Batch History Functionality

An operator can list the batch history of an output product in order to be able to trace the manufacturing process of a batch for analyses.

Batch Tracing Functionality

An operator can use the batch tracing functionality to verify through which machines/operations the throughput batch was produced. The operator thus sees the material's route through production.